

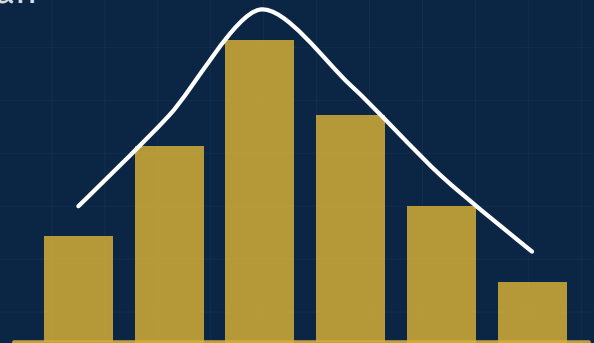


DR ESLAM AHMED
eliteigcse.com

CHAPTER 6 • STATISTICS & PROBABILITY • TOPIC 1

Statistics

Averages, spread and tables. The mean always comes back to the total; the median is the middle; the interquartile range is the middle half.



5

VARIANTS

58

PAST-PAPER PARTS

47

PRACTICE QUESTIONS

YOU WILL MASTER

V1 Averages & range from a list

V2 Mean problems: totals

V3 Grouped frequency tables

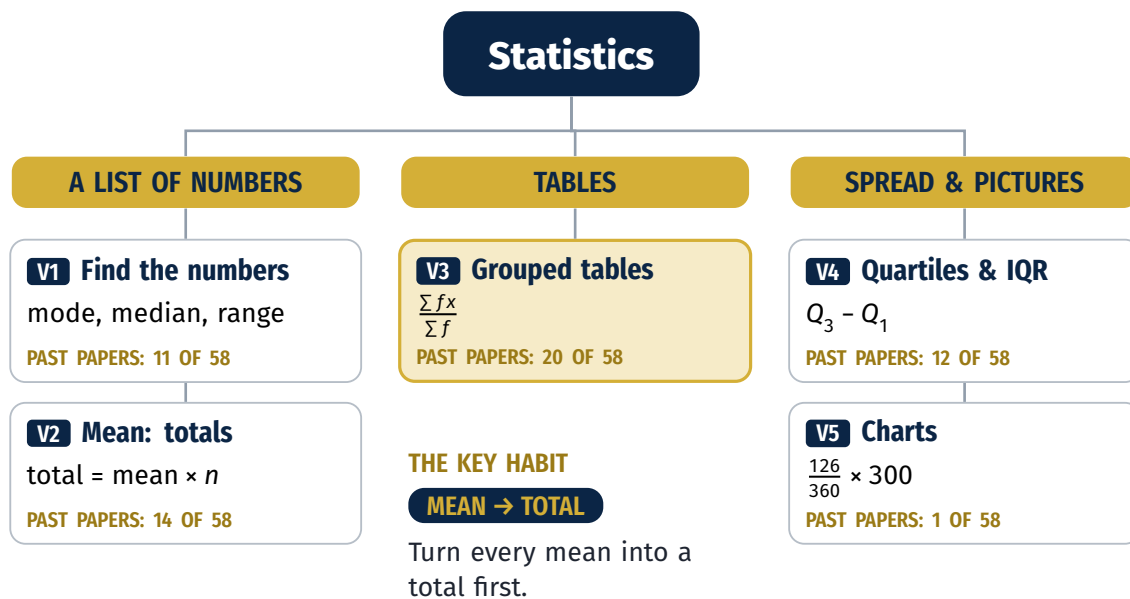
V4 Quartiles & IQR

V5 Charts

TOPIC MAP • EVERY VARIANT ON ONE PAGE

Statistics at a glance



Your notes


★ How often does each variant come up?

48 past-paper questions (Jan 2020 → Nov 2025) hold 58 parts; 47 practice parts — grouped by type.

	PAST PAPERS (OUT OF 58)	PRACTICE PARTS
V1 Averages, range from a list	11	9
V2 Mean problems: totals	14	11
V3 Grouped frequency tables	20	17
V4 Quartiles, IQR	12	5
V5 Charts	1	5

WHAT THIS TELLS YOU The biggest topic of all: about 4 parts on every paper, mostly Q1–Q14 (2–5 marks). Grouped tables, mean-by-totals and IQR are near-certain marks — learn the three methods.

BEFORE YOU START

The big idea



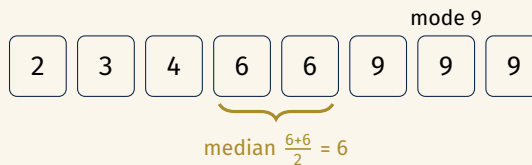
Put the numbers in order first.

Mean = $\frac{\text{total}}{\text{how many}}$. **Median**: the middle

value. **Mode**: the most common.

Range: largest – smallest.

IQR = $Q_3 - Q_1$: the spread of the middle half — it ignores extreme values.



The averages

Mean

$$\frac{\sum x}{n}$$

Total ÷ how many.

Median

$$\frac{n+1}{2} \text{th}$$

Middle value (in order).

Mode

most common

Can be more than one.

Range

max – min

Spread of all the data.



Spread

Lower quartile

$$\frac{n+1}{4} \text{th}$$

Quarter of the way.

Upper quartile

$$\frac{3(n+1)}{4} \text{th}$$

Three quarters of the way.

Compare

average, spread

Higher median; smaller IQR = more consistent.

**CALCULATOR FIRST, THEN CHECK**

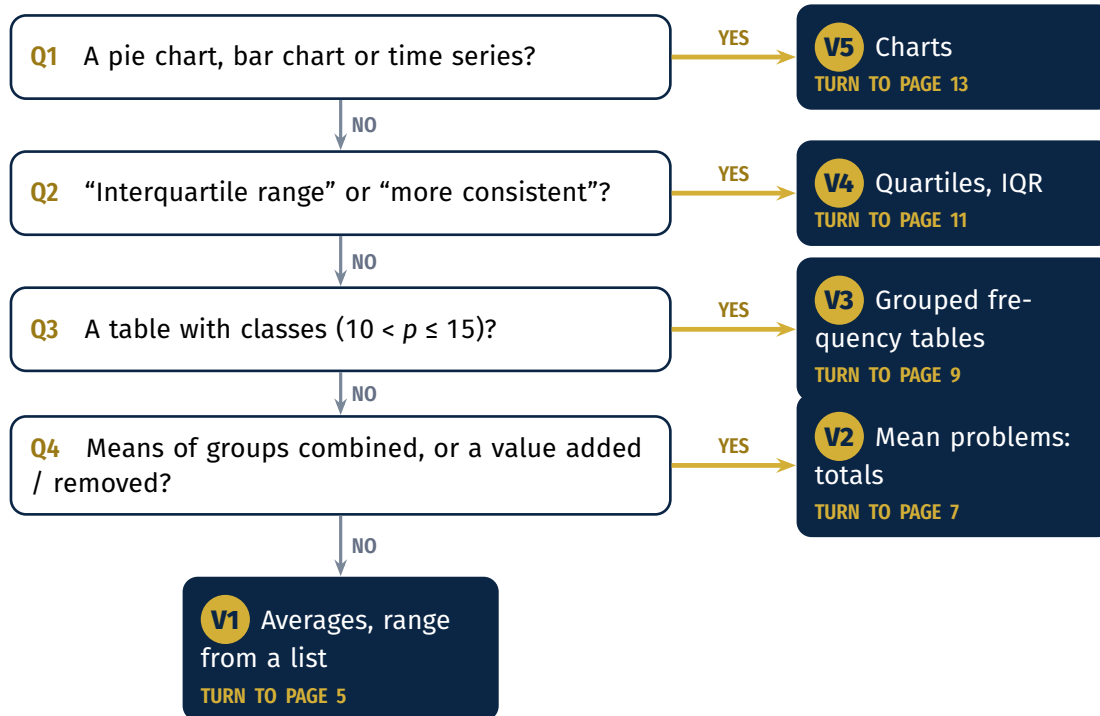
Calculator statistics mode (1-VAR) gives the mean of a list or a table — use it to check the mean.

COUNT / RECOGNISE, THEN CHOOSE

Which variant am I?

 Your notes

Read the question, then walk down the ladder. The first **YES** tells you the variant – and the page to turn to.



COLOUR CODE IN THE WORKING

Blue = letter terms $4x$, $-3y$

Orange = plain numbers 17 , -13

STEP 1 **STEP 2** **STEP 3** **STEP 4** = which step of "How do I do it?" each line is.

ON EVERY VARIANT SPREAD

- | | | | |
|---|-----------------|---|-----------------------|
|  | What is it? |  | When you see → you do |
|  | How do I do it? |  | Watch out |
|  | Worked example |  | Exam tip |
|  | Check yourself |  | Calculator check |

V1

VARIANT 1

Find the Numbers: Mode, Median, Range

Use each fact on its own: mode → repeats,
median → middle, range → ends, mean →
total.

SEEN IN PAST PAPERS

11 / 58

PRACTICE QUESTIONS: 9



Your notes

Came up in: Oct/Nov 2020 P1H Q4 • Jan 2021 P1HR Q2 • Jan 2021 P2H Q2 • Jan 2022 P1HR Q5 • Jan 2022 P1H Q5(a) • Jun 2022 P2HR Q3 • Jun 2022 P2H Q1 • Jun 2024 P2HR Q3 • Jun 2024 P2H Q1 • Jun 2025 P2H Q1 • Nov 2025 P1H Q4

? What is it?

In plain words

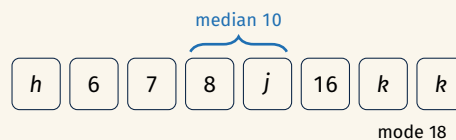
Missing numbers with clues. Write the list **in order** with gaps, then use one clue at a time:

Mode — which number repeats.

Median — the middle (two middles: their average). **Range** — the two ends. **Mean** — the total.

IT LOOKS LIKE

h 6 7 8 j 16 k k



How do I do it?

1 Order

Write the list in order with blanks.

2 Mode

The repeated number.

3 Median

Middle value(s).

4 Range, mean

Ends; total = mean $\times$ n .

ONE CLUE AT A TIME Write each clue as an equation.

V1 Find the Numbers: Mode, Median, Range

VARIANT 1 • SEE IT DONE, THEN TRY

 Your notes
WHEN YOU SEE → YOU DO

"The median of the eight numbers is 10" → **Average of the 4th and 5th = 10**

"The mode ... is 18" → **18 appears most**

"a mean of 9" with 6 numbers → **Total 54**

WORKED EXAMPLE **PAST PAPER**

June 2024 • Paper 2H • Q1 • 3 marks

Eight numbers in order: $h, 6, 7, 8, j, 16, k, k$. The median is 10, the mode is 18 and the range is 13. Find h, j and k .

STEP 1 $\frac{8+j}{2} = 10 \Rightarrow j = 12$ ◀ median: 4th and 5th

STEP 2 $k = 18$ ◀ mode

STEP 3 $18 - h = 13$ ◀ range

STEP 4 $h = 5$ ◀ solve

ANSWER $h = 5, j = 12, k = 18$

✓ 5, 6, 7, 8, 12, 16, 18, 18 ✓

WATCH OUT

✗ median = 4th value

✓ average of 4th and 5th

An even count has two middles.

EXAM TIP

Write the final list out and check every clue.

Check yourself

calculator first • answers on page 15

1.1 Jan 2021 P1HR

Whole numbers a, a, b, c ($a < b < c$): mode 7, median 8.5, mean 9.

Answer:

1.2 Nov 2025 P1H

In order: $e, f, 7, g, 12, 17$. Range 15, median 8.5, mean 9.

Answer:

1.3 Oct/Nov 2020 P1H

In order: 4, 7, x , 10, y, y . Median 9, mean 11.

Answer:

V2

VARIANT 2

Mean Problems: Work with Totals

Mean $\times$ number = total; add or subtract totals, never means.

SEEN IN PAST PAPERS

14 / 58

PRACTICE QUESTIONS: 11


Your notes

Came up in: Jan 2020 P1HR Q8 • Jan 2021 P2H Q7 • May 2021 P1H Q10 • Jan 2022 P2HR Q8 • Jun 2022 P1HR Q3 • Jun 2022 P2H Q12 • Jun 2023 P2HR Q10 • Jun 2023 P2H Q10 • Nov 2023 P2H Q8 • May 2024 P1HR Q1 • Jun 2024 P2H Q12 • Nov 2024 P2H Q8 • Jun 2025 P2HR Q8

? What is it?

In plain words

Means can't be added. **Totals** can.
Turn every mean into a total: total = mean $\times$ how many.

Then add or subtract totals, and divide at the end if a mean is asked.

$5 \times 104 = 520$	Sat, Sun
----------------------	----------

$7 \times 127 = 889$

IT LOOKS LIKE

5 days mean 104, 7 days mean 127

How do I do it?

1 Totals

Each mean $\times$ its count.

2 Combine

Add the groups, or take a part away.

3 Missing

The difference is the missing value or total.

4 Mean?

Divide by the new count if asked.

MEAN $\rightarrow$ TOTAL Add totals, not means.

V2 Mean Problems: Work with Totals

VARIANT 2 • SEE IT DONE, THEN TRY

 Your notes
WHEN YOU SEE → YOU DO

"A sixth cocoa pod is put in; the mean becomes 401" → $6 \times 401 - 5 \times 398$

"mean mark ... to be at least 77" → **Least total needed – total so far**

Two groups with their own means → **Total of each, then combine**

WORKED EXAMPLE PAST PAPER

June 2023 • Paper 2HR • Q10 • 3 marks

Mon–Fri Gemara's mean distance was 104 km; for all 7 days it was 127 km. On Saturday he drove 132 km. How far did he drive on Sunday?

STEP 1 $5 \times 104 = 520$ ◀ Mon–Fri total

STEP 2 $7 \times 127 = 889$ ◀ week total

STEP 3 $889 - 520 - 132$ ◀ take away the known days

STEP 4 $= 237$ ◀ Sunday

ANSWER 237 km✓ $520 + 132 + 237 = 889$ ✓**WATCH OUT**

✗ $\frac{104+127}{2}$

✓ totals: $889 - 520 - 132$

Means of different-sized groups can't be averaged.

EXAM TIP

Write "total =" lines — each one is a method mark.

Check yourself

calculator first • answers on page 15

2.1 Jan 2022 P2HR

22 Siberian tigers, mean 260 kg; 28 Bengal tigers; all 50 mean 218 kg. Mean of the Bengal tigers?

Answer:

2.2 May 2021 P1H

Mean of 5 tests is 74. Least mark in a 6th test for a mean of at least 77?

Answer:

2.3 Jun 2023 P2H

A: 11 rounds mean 17. B: 9 rounds mean 18. One more round each: both means 18.5. How many more did A score?

Answer:

V3

VARIANT 3 MOST COMMON IN THE EXAM

Grouped Frequency Tables

Midpoint $\times$ frequency; total $\div$ total frequency. Modal class has the highest frequency.

SEEN IN PAST PAPERS

20 / 58

PRACTICE QUESTIONS: 17

 Your notes

Came up in: Jan 2020 P1H Q4 • Jan 2020 P2HR Q2 • Oct/Nov 2020 P1HR Q4 • Jan 2021 P1H Q2 • Jan 2021 P2HR Q10 • Nov 2021 P2H Q3 • Jan 2022 P2H Q9 • May 2023 P1H Q2 • Nov 2023 P2H Q1 • Nov 2024 P1H Q1 • Jun 2025 P1H Q1 • Nov 2025 P1H Q1

? What is it?

In plain words

We don't know the exact values, so use each class's **midpoint**. Add a column: **midpoint $\times$ frequency**. Its total $\div$ the total frequency = **estimated mean**.

"Estimate the total" — stop before dividing.

IT LOOKS LIKE

10 < p ≤ 15

modal class

estimate

class	f	mid	fx
10 < p ≤ 15	18	12.5	225
15 < p ≤ 20	16	17.5	280
20 < p ≤ 25	14	22.5	315
25 < p ≤ 30	8	27.5	220
30 < p ≤ 35	4	32.5	130
	60		1170

How do I do it?

1 Midpoints

Half way across each class.

2 fx

Midpoint $\times$ frequency.

3 Totals

Σf and Σfx .

4 Divide

$\frac{\Sigma fx}{\Sigma f}$ (or stop for a total).

MIDPOINT $\times$ FREQUENCY The most common variant here.

V3 Grouped Frequency Tables

VARIANT 3 · SEE IT DONE, THEN TRY



Your notes
 WHEN YOU SEE → YOU DO

“Write down the modal class” → **Class with the highest frequency**

“Work out an estimate for the mean” → $\frac{\sum fx}{\sum f}$ **with midpoints**

“the estimated mean was 4.25 ... find x ” → **Equation with x in f**

 WORKED EXAMPLE **PAST PAPER**

November 2024 · Paper 1H · Q1 · 5 marks

Hourly pay of 60 workers: $10 < p \leq 15$: 18, $15 < p \leq 20$: 16, $20 < p \leq 25$: 14, $25 < p \leq 30$: 8, $30 < p \leq 35$: 4. (a) Modal class. (b) Estimate the mean.

STEP 1 $10 < p \leq 15$ ◀ (a) highest frequency (18)

STEP 2 midpoints 12.5, 17.5, 22.5, 27.5, 32.5 ◀ middle of each class

STEP 3 $\sum fx = 225 + 280 + 315 + 220 + 130 = 1170$ ◀ multiply and add

STEP 4 $1170 \div 60 = 19.5$ ◀ divide

ANSWER (a) $10 < p \leq 15$ (b) \$19.50 ✓ inside the range 10–35 ✓

 WATCH OUT

✗ dividing by 5 (the number of classes)

✓ divide by 60 (total frequency)

It is the mean of 60 workers.

 EXAM TIP

“An estimate” — because midpoints are used. Write the fx column out in full.

 Check yourself

calculator first · answers on page 15

3.1 Jan 2020 P1H

40 babies: $2 < w \leq 3$: 12,
 $3 < w \leq 4$: 16, $4 < w \leq 5$:
 9, $5 < w \leq 6$: 2, $6 < w \leq 7$:
 1. Estimate the mean.

Answer:

3.2 Jan 2021 P2HR

Distances: 0–2: x , 2–4:
 11, 4–6: 8, 6–8: 6, 8–10:
 5. Estimated mean 4.25
 km. Find x .

Answer:

3.3 Nov 2023 P2H

50 calls in 5-minute
 classes from 0 to 30: 8,
 2, 6, 4, 12, 18. Estimate
 the total length.

Answer:

V4

VARIANT 4

Quartiles & Interquartile Range

In order; Q_1 at $\frac{n+1}{4}$, Q_3 at $\frac{3(n+1)}{4}$; $IQR = Q_3 - Q_1$.

SEEN IN PAST PAPERS

12 / 58

PRACTICE QUESTIONS: 5



Your notes

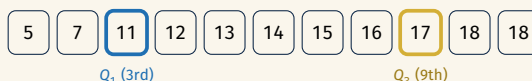
Came up in: Jan 2020 P2HR Q14 • Jan 2021 P2H Q12 • May 2021 P1H Q14 • Jan 2022 P2H Q15 • Jun 2022 P1HR Q13 • May 2023 P1H Q10 • Jun 2023 P2HR Q14 • Nov 2023 P2H Q13 • May 2024 P1HR Q10 • Jun 2025 P1HR Q13 • Nov 2025 P1H Q16

? What is it?

In plain words

Put the values **in order**. With n values: Q_1 is the $\frac{n+1}{4}$ th, Q_3 the $\frac{3(n+1)}{4}$ th.

$IQR = Q_3 - Q_1$. A **smaller** IQR means **more consistent**.



IT LOOKS LIKE

11 values

15 values

How do I do it?

1 Order

Smallest to largest; count n .

2 Q_1

 $\frac{n+1}{4}$ th value.

3 Q_3

 $\frac{3(n+1)}{4}$ th value.

4 IQR

 $Q_3 - Q_1$; compare if asked.

ORDER FIRST

11 values: 3rd and 9th. 15 values: 4th and 12th.

V4 Quartiles & Interquartile Range

VARIANT 4 • SEE IT DONE, THEN TRY


 Your notes
 WHEN YOU SEE → YOU DO

"Find the interquartile range" → **Order; $Q_3 - Q_1$**

"Who had the more consistent scores?" → **Smaller IQR, with a reason**

Values not in order → **Sort them first**

**WORKED EXAMPLE** **PAST PAPER**

January 2020 • Paper 2HR • Q14 • 3 marks

Max's 11 spelling-test marks: 18, 5, 7, 12, 11, 18, 15, 16, 17, 13, 14. Find the interquartile range.

STEP 1 5, 7, 11, 12, 13, 14, 15, 16, 17, 18, 18 ◀ in order, $n = 11$

STEP 2 $Q_1 = 3\text{rd} = 11$ ◀ $\frac{12}{4} = 3$

STEP 3 $Q_3 = 9\text{th} = 17$ ◀ $\frac{36}{4} = 9$

STEP 4 $17 - 11 = 6$ ◀ IQR

ANSWER 6

✓ middle half spans 11 to 17 ✓

**WATCH OUT**

× quartiles from the unsorted list

✓ sort first

Positions only work in order.

**EXAM TIP**

"More consistent": say "smaller interquartile range" with both numbers.

**Check yourself**

calculator first • answers on page 15

4.1

Jun 2023 P2HR

Akari: 25, 20, 28, 27, 26,
22, 23, 29, 20, 29, 26.
Find her IQR. Machi's
IQR is 9. Who is more
consistent?

Answer:

4.2

Jan 2021 P2H

15 waiting times: 2, 3, 3,
4, 5, 6, 6, 8, 9, 10, 11, 13,
14, 15, 18. IQR?

Answer:

4.3

Nov 2025 P1H

15 values: 1, 1, 3, 4, 6, 7,
8, 9, 10, 12, 15, 25, 30, 37,
50. IQR?

Answer:

V5

VARIANT 5

Charts: Pie, Bar & Time Series

Pie: angle = $\frac{\text{frequency}}{\text{total}} \times 360^\circ$, and back again.

SEEN IN PAST PAPERS

1 / 58

PRACTICE QUESTIONS: 5



Your notes

Came up in: Nov 2024 P2H Q10

? What is it?

In plain words

In a pie chart the **whole circle** (360°) is the **whole total**. Each sector's share: $\frac{\text{angle}}{360}$.

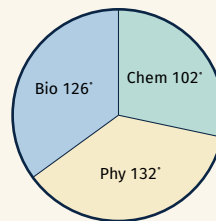
Number in a sector = $\frac{\text{angle}}{360} \times \text{total}$.

Angle = $\frac{\text{number}}{\text{total}} \times 360$.

IT LOOKS LIKE

126° of 300

30 of 72



How do I do it?

1 Share

 $\frac{\text{angle}}{360}$ or $\frac{\text{number}}{\text{total}}$.

2 Convert

 $\times \text{total}$, or $\times 360$.

3 Compare

Same units before comparing groups.

4 Answer

The difference or value asked.

360° = THE WHOLE Angles are shares.

V5 Charts: Pie, Bar & Time Series

VARIANT 5 · SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO

"There are 300 students in year 10"
with a pie chart

$$\rightarrow \frac{\text{angle}}{360} \times 300$$

A table with x and a total

→ **Add the expressions = total; solve**

"Work out the angle for ..."

$$\rightarrow \frac{\text{number}}{\text{total}} \times 360$$

WORKED EXAMPLE PAST PAPER

November 2024 · Paper 2H · Q10 · 5 marks

Year 10 (300 students): Biology 126° on a pie chart. Year 11 (320 students): Biology $3x + 6$, Chemistry $5x + 8$, Physics $7x - 9$. How many more year 10 than year 11 students chose Biology?

STEP 1 $\frac{126}{360} \times 300 = 105$

◀ year 10 Biology

STEP 2 $15x + 5 = 320 \Rightarrow x = 21$

◀ year 11 total

STEP 3 $3(21) + 6 = 69$

◀ year 11 Biology

STEP 4 $105 - 69 = 36$

◀ difference

ANSWER 36 more

$$\checkmark 69 + 113 + 138 = 320 \checkmark$$

WATCH OUT

× comparing 126° with 69

✓ convert the angle to people first

Angles and numbers are different units.

EXAM TIP

Check the year-11 numbers add up to 320.

Check yourself

calculator first · answers on page 15

5.1

Dr Eslam

72 people; 30 chose red.
Angle for red in a pie chart?

Answer:

5.2

Dr Eslam

A 90° sector represents
45 people. How many
people in total?

Answer:

5.3

Practice · Medium

Monthly values 110, 84,
78, 94, 90, 120. Find the
mean.

Answer:

CHECK YOURSELF

Answers



Your notes

Every answer below was checked in Python (check_answers.py).

1.1 $a = 7, b = 10, c = 12$

1.2 $e = 2, f = 6, g = 10$

1.3 $x = 8, y = 18.5$

2.1 185 kg

2.2 92

2.3 12

3.1 3.6 kg

3.2 $x = 10$

3.3 945 minutes

4.1 IQR 6; Akari

4.2 9 minutes

4.3 21

5.1 150°

5.2 180

5.3 96

THE WHOLE TOPIC ON ONE PAGE

Statistics cheat sheet



Your notes
V1 Find numbers 11/58

$$\frac{8+j}{2} = 10$$

One clue at a time.

V2 Totals 14/58

$$7 \times 127 - 5 \times 104$$

Mean → total.

V3 Grouped 20/58

$$\frac{\sum fx}{\sum f}$$

Midpoints; modal class.

V4 IQR 12/58

$$Q_3 - Q_1$$

Order first; smaller = consistent.

V5 Charts 1/58

$$\frac{126}{360} \times 300$$

360° is the whole.



TOP 3 LOST MARKS 1. averaging means × 2. dividing by the number of classes × 3. quartiles from an unsorted list ×

BOTTOM LINE Order first · mean → total · midpoint × frequency · IQR
= $Q_3 - Q_1$.